NOLA Assessment Navigator, Part 2: Data Exploration, Property Lookup, and Chatbot

Live chatbot: https://claude.ai/artifact/PMnTiHc5EGih7pzGky6bWB. Find a property by address or parcel ID, confirm the record, then ask about it. **Data:** `Orleans_Parish_2026_LTC_Expanded_Real_Estate_Roll.csv`: Louisiana Tax Commission 2026 roll, 163,135 parcels, 115 columns. **Code and full docs:** https://github.com/ethanpzuckerman/ethanpzuckerman.github.io/tree/claude/elegant-brahmagupta-747i4y/nola-assessment-navigator
Code: `pipeline/` (analysis and export), `address.js` + `lookup.js` (matching), `property.js` (property answers), `tests/` (all checks below are rerunnable).

Where each requirement is answered:

Assignment item	Where
1. EDA: 3–5 charts with a paragraph each, plus a finding verified independently	§1: Figures 1–5, then "Verifying a finding independently"
2. Data dictionary, plus what we couldn't figure out	§2, and all 115 fields in Appendix A (`docs/data-dictionary.md`)
3. Address matching tested on 10+ addresses: how many matched, and why any failed	§3, and all 19 results in Appendix B (`docs/address-matching-results.md`)
4. One outside source pulled by API, saved and documented	§4: Census ACS by tract; `data/outside/`, `docs/outside-data.md`
5. 10 test questions: expected answer vs. what the chatbot did	§5, and screenshots of every answer in Appendix C
Live link to the current chatbot	Top of this page
Compare 2025 and 2026 by a consistent ID	§4b: 2025 isn't available, so 2026 vs 2027 by parcel ID
Property lookup: address variations, multiple matches, missing records	§3, and the live chatbot's "Find your property" box
Property records connected to the chatbot (data year, source, missing info, "can't answer")	§5; every property answer labels what was recorded, calculated and explained
Evaluate outside data and matching limits	§4
CLAUDE.md updated as we learned	`CLAUDE.md` §3 (Data), §4 (rules 7–8), §6–8

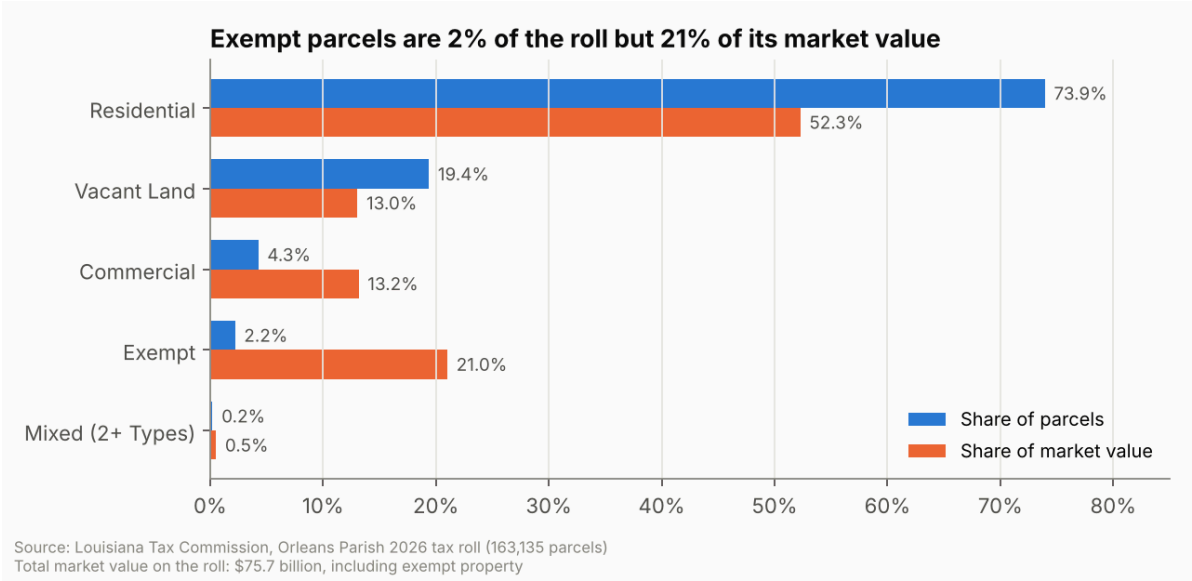
1. Exploratory data analysis

The agent ran `pipeline/eda.py`, which also writes every number below to `docs/eda/eda_numbers.json`. First it checked that the roll's own arithmetic holds. On **all 163,135 rows**:

- market value = land + buildings

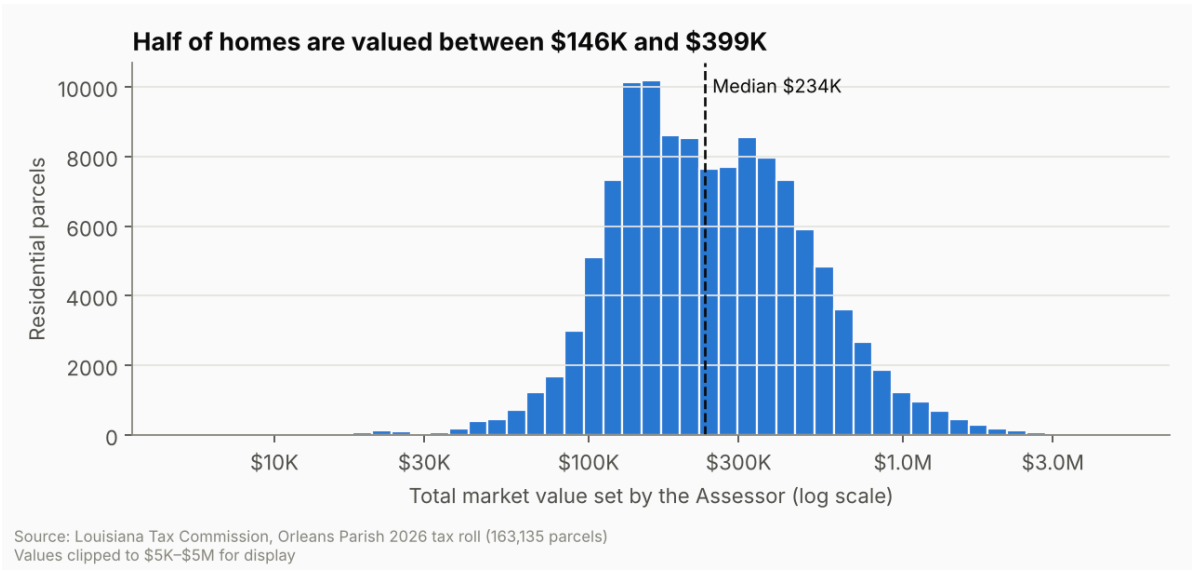
- assessed value = land + buildings (assessed)
- taxable value = assessed − homestead − veteran − first responder − other exemptions, never below $0
- Parcel ID = Tax Bill #
- the homestead exemption never exceeds $7,500

Figure 1: What's on the roll, by count and by value (`docs/eda/fig1_share_parcels_value.png`)



Residential parcels are 74% of the roll but 52% of its $75.7 billion market value. **Exempt parcels are only 2.2% of parcels but 21% of the value.** These are large institutional properties: City Park ($1.39 billion), the LSU and VA medical centers (about $1 billion each), Tulane, the Convention Center, the Superdome and the Port. None of that value is taxed. Commercial parcels are 4% of the count and 13% of the value. 14,841 parcels have a taxable value of $0: exempt property, vacant land with an "other" exemption, and 4,197 homes whose exemptions cover the whole assessed value.

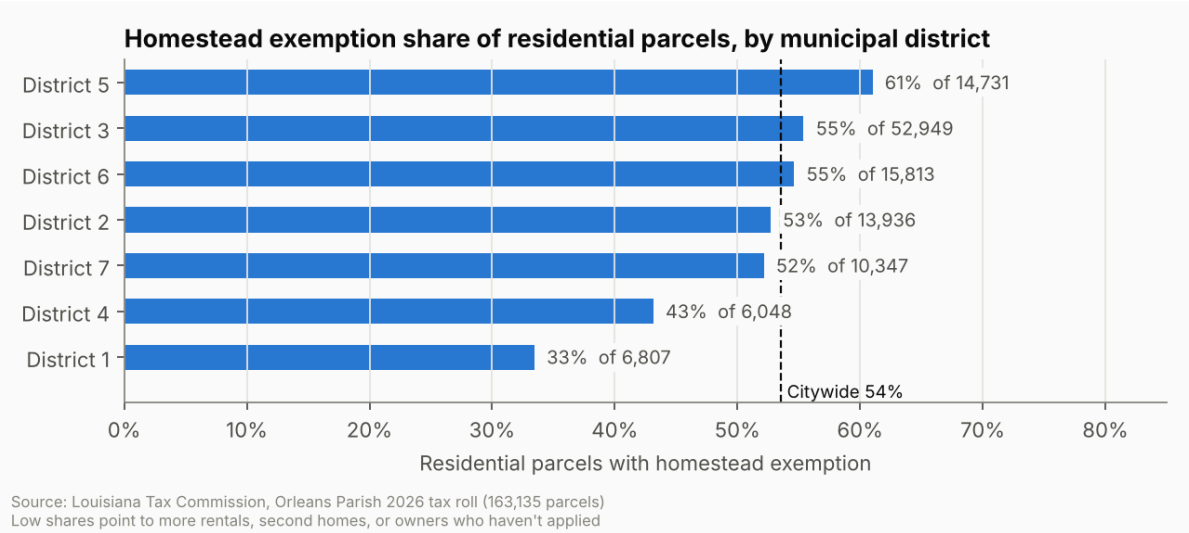
Figure 2: How residential values are distributed (`docs/eda/fig2_residential_values.png`)



The median home is valued at **$233,600**, and half of homes fall between **$145,900 and $398,550**. The 10% at the top are above $611,800. These are the Assessor's mass-appraisal values as of

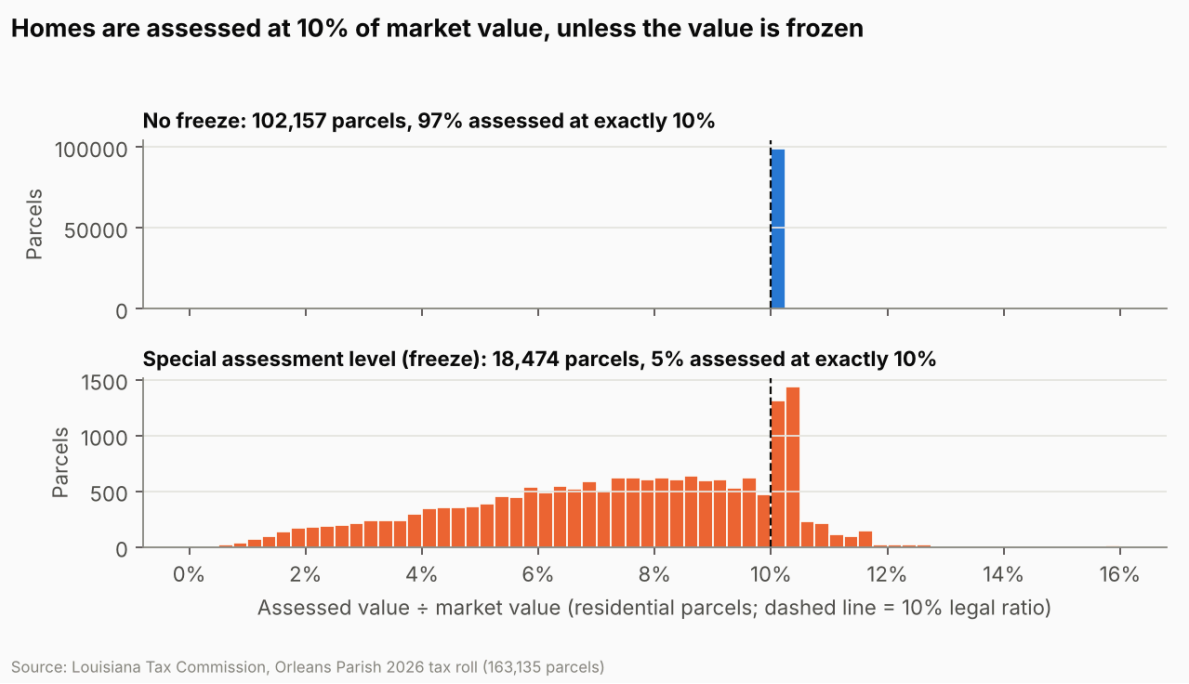
01/01/2023, not current sale prices.

Figure 3: Homestead exemption by district (`docs/eda/fig3_homestead_by_district.png`)



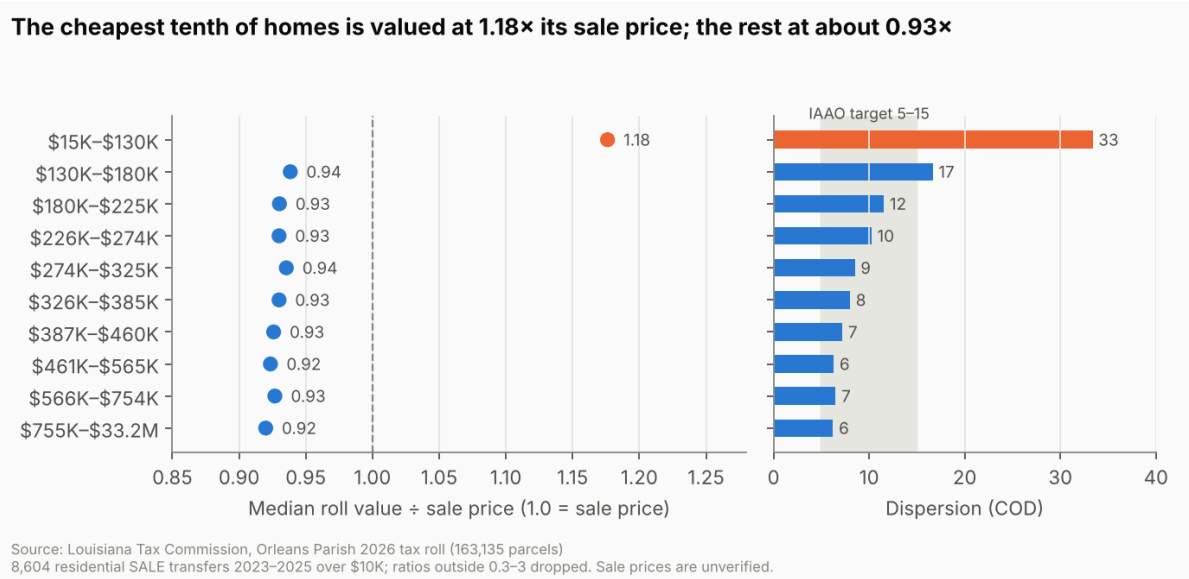
Citywide, **54% of residential parcels** have a homestead exemption. That ranges from **61% in District 5** to **33% in District 1**. A low share can reflect more rentals, second homes and short-term rentals, or owners who live there but haven't applied. That makes it a useful map for homestead outreach.

Figure 4: The 10% rule, and the freeze exception (`docs/eda/fig4_assessment_ratio.png`)



For homes without a freeze, **97% are assessed at exactly 10% of market value**, the legal ratio in La. Const. Art. VII §18. The 18,474 homes with a **special assessment level (freeze)** are spread below that, with a median of **7.6%**, because their assessed value was locked when the owner qualified while market value rose. This one chart shows two rules from the education outline working in the data.

Figure 5: Are cheaper homes over-assessed? (the client's question)
(`docs/eda/fig5_sales_ratio_by_price.png`)



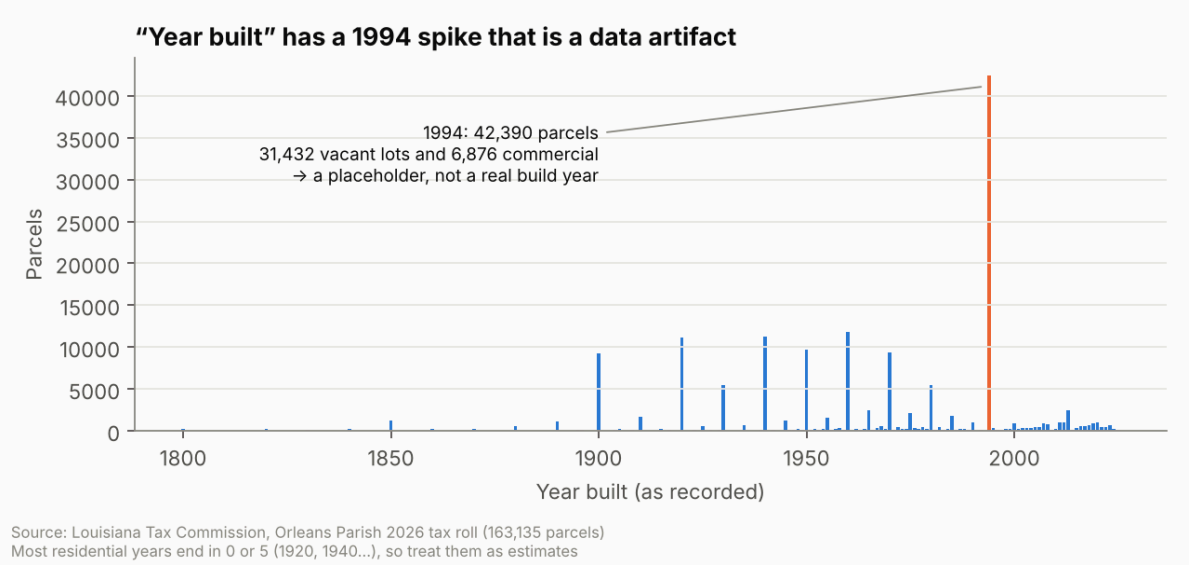
For 8,604 residential sales from 2023–2025, we compare the roll's market value with the sale price. The middle and top of the market sit at **0.92–0.94× the sale price**. The **cheapest tenth ($15K–$130K) sits at 1.18×**, with dispersion (COD 33) more than twice the IAAO target band of 5–15. Taken at face value, owners of the least expensive homes are assessed at more than they sold for, while everyone else is assessed below. That's the regressivity concern the project's client raised.

Caveats before anyone acts on it:

- None of the roll's sale prices are marked verified.
- The cheapest tenth likely includes distressed and non-market sales. Including every deed type (sheriff's sales, successions and other non-SALE transfers) raises that tenth to 1.41×.
- Values are as of 01/01/2023, so later sales include appreciation.
- With 2021–22 sales, from before the valuation date (and with slightly different filters: SALE, SALW and CASH documents, prices of $25K or more), the cheapest tenth is at 1.05×. The gap is smaller there, but it points the same way.

A related pattern: 19% of these sales (40% of 2023 sales) carry a roll value at almost exactly **92% of the sale price**, and 15% at exactly 92% rounded to $100. That suggests values are reset when a home sells ("sales chasing"), which would also make sales-based accuracy measures look better than they are for homes that haven't sold.

Appendix figure A1: a data-quality warning (`docs/eda/figA1_year_built.png`)



"Year built" spikes at **1994: 42,390 parcels**, of which 31,432 are vacant lots and 6,876 commercial. That's a placeholder, not a building boom. Residential years are rounded too: 79% end in 0 or 5. The app hides 1994 for non-residential parcels and calls residential years approximate.

Verifying a finding independently

We rechecked the Figure 4 finding a second way, with Python's built-in `csv` reader instead of pandas, reading the raw file directly:

- **No freeze:** 99,289 of 102,157 homes at exactly 10% (**97.19%**)
- **Freeze:** 963 of 18,474 (**5.21%**)

Both match the EDA. We also traced one parcel by hand. For **1205 St Charles Av #607**, land $12,700 + building $133,000 = $145,700 market value. × 10% = $14,570 assessed, − $7,500 homestead = **$7,070 taxable**, exactly what the roll records.

To reproduce it in Excel (the file's 163,135 rows fit within Excel's limit): filter Occupancy Type = RESIDENTIAL and Freeze Code = 0, add a column `=Total Assessed Value / Total Market Value`, and count the rows equal to 0.1. The result should be 99,289 of 102,157.

Second check: replicating the class model EDA. The Oct. 28 class recording shows the instructor's AI-built EDA. We reran its two headline findings with our own code on the raw file (`pipeline/eda.py`, `sales_ratio()`):

Finding in the class model EDA	Our independent result
8,604 residential sales, 2023–2025, over $10K, ratios outside 0.3–3 dropped	**8,604**, which matches only when the filter is document type = SALE. That's an unstated filter we had to discover.
Homes under $130K valued at 1.18× price; every other tenth at 0.92–0.94	**1.18** for $15K–$130K; **0.92–0.94** for every other tenth
COD 33.4 in the cheapest tenth, falling to about 6 at the top	**33.4**, falling to **6.2**
About 1 in 5 sales at almost exactly 92% of price; 40% of 2023 sales	**19.4%** (1,672 sales); **40%** of 2023, 9% of 2024, 6% of 2025

Finding in the class model EDA	Our independent result
15% equal 92% of price rounded to $100	**15.3%**

Two models and two codebases agree, which makes the finding much more credible. Our extra checks are the caveats under Figure 5.

2. Data dictionary

Appendix A (`docs/data-dictionary.md`, generated by `pipeline/data_dictionary.py`) covers all 115 fields. Each one has a meaning, a status (Verified / Inferred / Unknown), its fill rate, and common values (hidden for owner and mailing fields).

What we couldn't figure out:

- **Code meanings:** what Freeze Code 1–4 each mean, the "other exemption" codes, the LTC subclass codes, and most of the 434 instrument-type codes.
- **The assessment date:** whether 01/01/2023 on every row is the 2024 reappraisal's valuation date.
- **Mislabeled fields:** "Living Square Feet" is actually a record count, not square footage.
- **Empty fields:** 39 are always blank, zero or "NOT AVAILABLE", including building size, bedrooms, bathrooms, condition, tax-sale status and special taxing district.

The chatbot says "I don't know" for all of these.

3. Address matching

Method (`address.js`, `lookup.js`): normalize the owner's input and the roll's addresses the same way. That covers:

- suffixes: Avenue/AV, Boulevard/BL, Highway/HW, Parkway/PW
- Saint/St and General/GEN
- MLK and O.C. Haley spellings
- the 2021 street renames
- units (Apt / # / Unit, or a bare unit as the roll writes it)
- half addresses (1/2 vs HF)
- city and ZIP tails

The matcher then looks up street + house number, falls back to close spellings, and accepts a 9-digit parcel ID.

Tested on 19 addresses (full table in **Appendix B**; `node tests/address_eval.js`): **19 of 19 behaved as expected**, with expected results written before running:

- **13 exact matches**, including:
  - "1205 St. Charles Avenue, Apt 607, New Orleans, LA 70130"
  - "2000 MLK Blvd"
  - "1050 Jefferson Davis Parkway" (now Norman C Francis)
  - "1518 1/2 Melpomene St"
  - a parcel ID

- **1 close match** to confirm: "1116 Tchoupitoulis St" (misspelled).
- **2 multiple matches**, as designed:
  - "1205 Saint Charles Ave" has 225 condo units, so the app asks for the unit number.
  - "701 S. Claiborne Ave" is 3 parcels, because **the roll drops N/S**. The app ranks the one whose legal description says "S CLAIBORNE" first.
- **3 not found, each with a reason:**
  - the house number isn't on that street
  - Metairie is outside Orleans Parish
  - no house number was given

Why matches fail or need a choice:

- The roll has no N/S direction.
- 1,110 addresses are shared by more than one parcel (condos and stacked records).
- 1,004 address strings have no usable street.
- Vacant lots often have no real street address.

4. Outside data

Source: U.S. Census Bureau **American Community Survey, 2024 5-year estimates (2020–2024)**, for all **184 Orleans census tracts** (371,853 residents). It covers:

- race and ethnicity (B03002)
- median household income
- owner vs. renter households
- median home value and median rent
- residents 65+
- vacant units

These are the same tables PolicyMap and Social Explorer show.

How it was pulled: by API with `pipeline/fetch_acs.py`, through **Census Reporter's public API**, which needs no key. The Census Bureau's own API now requires a free key; set `CENSUS_API_KEY` and the same script uses it instead.

Saved to: `data/outside/acs_orleans_tracts.csv`, with a margin of error for every estimate, and `.meta.json` (endpoint, release, tables, pull date).

Can it be matched to the roll? Tested (`docs/outside-data-matching.md`, `pipeline/geocode_sample.py`). The roll has no coordinates or tract, so we sent a random sample of **504 parcels** to the Census batch geocoder and joined the tract it returned to the ACS table.

	Share of the sample
Joined to an ACS tract, all parcels	**86%** (432 of 504)
Homes	**94%**
Commercial	81%
Vacant land	**57%**

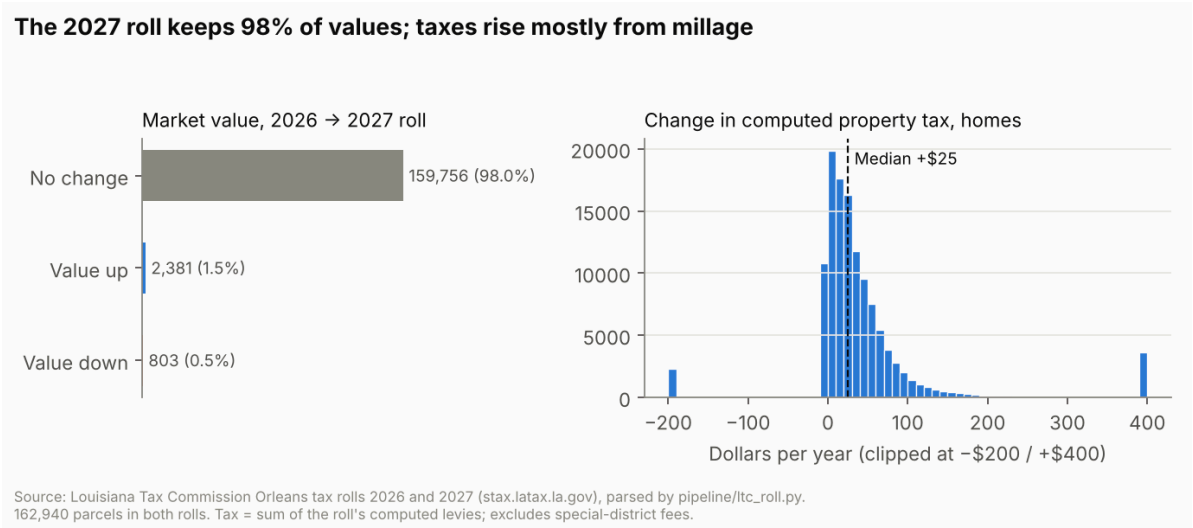
	Share of the sample
Ties: two possible locations, mostly N/S streets the roll can't tell apart	6%

Verdict: reliable enough for **neighborhood context on homes**, but not for every parcel. Vacant lots often have no mailable address. A parcel-ID join to the City's parcel GIS layer is the next thing to test. Tract figures describe a neighborhood, never an owner, so they stay out of property answers for now (CLAUDE.md §3 rule 7).

4b. Comparing years (2026 vs 2027)

The assignment asks for 2025 vs 2026 "where available." **The 2025 Orleans roll isn't available**: the course Box folder has only 2026, and the Louisiana Tax Commission's site keeps only its two most recent upload years. Those years hold the **2026 roll and the new 2027 roll** (filed Sept. 4, 2026), so we compared those, by parcel ID (`pipeline/ltc_roll.py`, `pipeline/compare_rolls.py`, full write-up in `docs/year-comparison.md`).

Parser check first: our parse of the raw 2026 tax commission files reproduces the course's Box file **exactly**. Market, land and taxable values match on all 163,135 parcels.



- **Parcels:** 162,940 are in both rolls. 195 drop out and 198 are new (splits, merges or renumbering), and these are reported, not guessed.
- **Values barely moved: 98% of parcels have exactly the same market value**, and every parcel is still dated 01/01/2023. Of the increases, 55% are on parcels with a recorded sale since 2025. The rest are likely construction, demolition or corrections (not checked).
- **Taxes moved because millage went up:** the median home's total rate went from **131.99 to 133.48 mills**. 84% of homes' computed tax rises, with a median increase of **$25**. The median homesteaded home goes from $1,965 to $2,032.
- **This contradicts our earlier sources,** which said the 2027 roll would carry a parish-wide reappraisal. The filed roll doesn't show one. The chatbot now says what the filed roll shows and that it doesn't know why. Ask the Assessor's Office (verification checklist).

Bonus from the same files: each parcel's millage rows give its **actual computed tax**, so the chatbot no longer estimates. The unverified 121.20-mill figure was too low; the real 2026 East Bank total is 131.99

mills. The files also confirm that homestead doesn't reduce the police and fire millages "without homestead exemption" (11.51 mills).

5. Ten test questions

Run in the real page with `node tests/chatbot_eval.js`. Expected answers come from the **raw CSV** (`tests/expected_from_raw.py`), not the app's own export. Screenshots of every answer are in **Appendix C**.

#	Property	Question	Expected (from the raw roll)	How it did
1	1205 St Charles #607	What is my property's market value?	$145,700 = $12,700 land + $133,000 buildings	✅ Exact, labeled Recorded with field names
2	1205 St Charles #607	How much tax will I pay?	Recorded tax $1,019.48 on the 2026 roll ($1,029.92 on 2027), summed from the raw millage rows	✅ Recorded amount, a calculated check ($14,570 × 131.99 − $904 homestead = $1,019) and the special-district caveat. *v1 used an unverified 121.20-mill estimate ($857); fixed once the millage files were available.*
3	423 Gravier #3A	Do I have a homestead exemption?	Yes, $7,500	✅ Plus the tax it removed on the 2026 roll ($904, recorded in the millage file)
4	1503 Tchoupitoulas	Why is my assessed value what it is?	$25,620 ÷ $325,000 = 7.9%, below 10% because of the freeze	✅ Explains the freeze
5	934 Melpomene	Why is my taxable value zero?	$19,810 − $7,500 − $12,310 = $0 taxable, yet the roll computes $228 (police/fire millages still apply)	✅ after a fix: v1 implied a $0 taxable value meant no tax
6	1770 Tchoupitoulas	How much tax will I pay?	$0, because the property is exempt (other exemption $659,150)	✅ after a fix: v1 said $0 without explaining why
7	1503 Tchoupitoulas	How did my value change from 2025?	Can't compare with 2025; 2026 → 2027: market $325,000 → $334,900, tax $2,478 → $2,505	✅ Real comparison by parcel ID; explains the freeze kept taxable value flat
8	1205 St Charles #607	How many square feet is my house?	Not in this roll	✅ "I don't know" plus where to look
9	1205 St Charles #607	Who owns this property?	Won't show names (privacy)	✅ after a fix: v1 routed "Who..." to the general contacts answer
10	(search)	701 S Claiborne Ave	3 records; S. Claiborne first	✅ after adding the legal-description ranking

Beyond the ten: `node tests/property_sweep.js` asks all 14 property question types about 305 random parcels of every kind (homestead, rental, commercial, vacant, exempt, frozen, veteran, multi-record),

4,270 answers, and flags broken output or a "matches the recorded amount" claim that isn't true. It found two bugs, both fixed:

- **Doubled mills:** on 866 parcels with several assessment records, the displayed rate counted each levy twice. The recorded taxes were always right. After the fix, every one of 150,793 parcels without other exemptions reconciles: assessed × mills − homestead = recorded tax, within $2.
- **Wrong routing:** "I think my value is wrong" was answered as a value lookup instead of with the review and appeal path.

The final sweep: **0 problems**.

Remaining limitations

- **No 2025 roll anywhere we can reach,** so "change from 2025" is answered with 2026 → 2027 and a clear note.
- **The 2027 reappraisal question is open:** the filed 2027 roll keeps 98% of values. Confirm with the Assessor.
- **Taxes exclude special-district fees** (DDD, security districts). They aren't in the state roll files.
- **Demographics aren't in property answers yet.** Matching works for 94% of homes but not for vacant land; a parcel-ID GIS join is next.
- **The chatbot is still keyword-based**, with no AI model behind it. Property answers are templates filled from the record, so they can't make up numbers, but they only cover the question types in `property.js`.
- **Code meanings are unknown** (freeze codes, exemption codes), so the bot says it doesn't know them.

Appendix A. Data dictionary (all 115 fields)

File: `Orleans_Parish_2026_LTC_Expanded_Real_Estate_Roll.csv` (Box, course data folder) **Origin:** Louisiana Tax Commission 2026 tax rolls for Orleans Parish, one archive per municipal district (`LTC Source URL`: https://stax.latax.la.gov/TaxRollsFolder), combined into one file. **Size:** 163,135 rows (one per parcel), 115 columns, UTF-8 with BOM. **Generated by:** `pipeline/data_dictionary.py`. Fill % counts values other than blank, `0` and `NOT AVAILABLE`.

Status key:

- **Verified:** checked against the data, e.g. an identity that holds on every row.
- **Inferred:** our best reading of the pattern; not confirmed by the Assessor or the LTC.
- **Unknown:** we could not work out what it means.

Examples are hidden for owner and mailing fields (privacy).

Field	Status	Fill %	Distinct	Most common values	Meaning
Tax Year	Verified	100%	1	`2026`	Tax year of the roll. Always 2026 (bills due early 2026).
Municipal District	Verified	100%	7	`3`, `5`, `6`	Orleans Parish assessment district 1–7. Each district maps to specific wards (see Ward(s)).
FIPS Code	Inferred	100%	7	`22173`, `22175`, `22176`	LTC code for the district's tax roll (22171–22177, one per municipal district), not a county FIPS code.
Tax Bill #	Verified	100%	163,135	`101100224`, `101100227`, `101100301`	Tax bill number. Identical to Parcel ID on every row.
Parcel ID(s)	Verified	100%	163,135	`101100224`, `101100227`, `101100301`	Unique parcel identifier (163,135 unique values). Use this to join years and other data.
Property Address(es)	Verified	100%	162,007	`210 BARONNE ST`, `5500 CHEF MENTEUR HW`, `14601 OLD GENTILLY HW`	Situs (property) address, roll format: no N/S direction, short suffixes (AV, BL, HW, PW, CR, WY), units after the suffix ("ST CHARLES AV 607"), "HF" for half addresses. 820 parcels list several addresses joined with " | ".
Owner(s)	Verified	100%	124,489	(private)	Owner name(s) as recorded. Not used in the app (privacy).
Primary Owner(s)	Verified	100%	124,426	(private)	First-listed owner. Not used in the app.
Assessment Status(es)	Inferred	100%	2	`AC`, `EX`	AC = active/taxable; EX = exempt (10,109 parcels, all vacant land or exempt occupancy).
Ward(s)	Verified	100%	17	`9W`, `13`, `7W`	Ward number (01–17, with 7W/8W/9W). Wards nest inside municipal districts.
Subdivision(s)	Verified	99%	611	`I-510 WEST`, `LOWER 9TH WARD 2`, `INNER CITY 7TH WARD`	Subdivision or area name used by the Assessor (611 values). Blank on 1,831 parcels.
Block(s)	Unknown	0%	0		Always blank in this file.
Lot(s)	Unknown	0%	0		Always blank in this file. Square/lot appear inside Legal Description instead.

Field	Status	Fill %	Distinct	Most common values	Meaning
Occupancy Type(s)	Verified	100%	12	`RESIDENTIAL`, `VACANT LAND`, `COMMERCIAL`	Use class: RESIDENTIAL, VACANT LAND, COMMERCIAL, EXEMPT; multi-record parcels list several joined by " | ".
Vacant Lot Indicator(s)	Verified	100%	4	`N`, `Y`, `Y | N`	Y if the parcel is vacant land. Matches Occupancy = VACANT LAND exactly (31,608).
Transfer Date(s)	Inferred	100%	14,663	`12/31/1994`, `10/14/2024`, `10/24/2022`	Date of the last recorded transfer. 12/31/1994 on 8,112 parcels looks like a placeholder for "before records were digitized".
Purchase Price(s)	Inferred	69%	14,417	`0`, `10`, `100`	Price on the last transfer document. 0 on 50,933 parcels and nominal amounts ($10, $100) are common, and none are marked verified. Usable for a sales-ratio study only with filters (document type SALE, recent years, price over $10K, outlier ratios dropped). See Figure 5 in docs/week2-report.md.
Verified Sale Indicator(s)	Inferred	100%	1	`N`	Whether the sale was verified as arm's-length. N on every row, so none are verified in this file.
Conveyance Book(s)	Unknown	0%	0		Always blank.
Conveyance Folio(s)	Unknown	0%	0		Always blank.
Instrument Number(s)	Unknown	0%	0		Always blank.
Instrument Type(s)	Inferred	95%	434	`SALE`, `QUIT`, `SUCC`	Type of the last transfer document: SALE, QUIT (quitclaim), SUCC (succession), SALW (sale with warranty?), etc. 434 codes; many are not explained.
Lender ID(s)	Unknown	0%	0		Always blank.
Tax Sale Indicator(s)	Unknown	0%	0		Always blank. Tax-sale status is not in this file.
Tax Sale Date(s)	Unknown	0%	0		Always blank.
Assessment Date(s)	Inferred	100%	1	`01/01/2023`	01/01/2023 on every row. Likely the valuation date of the 2024 reappraisal that 2026 values carry forward from.
Subdivision Phase(s)	Unknown	0%	0		Always blank.
Legal Description(s)	Verified	100%	161,015	`SQ 274 JULIA ST, S RAMPART`, `SQ 227 CARONDELET ST AND CO`, `SQUARE 14X CANAL ST TO POYDR`	Legal description (square, lot, bounding streets). Useful to tell apart parcels that share an address.
Building Use(s)	Inferred	100%	4	`R`, `C`, `C | R`	R = residential, C = commercial building use.
Structure Number(s)	Unknown	0%	0		Always blank.
Auxiliary Square Feet	Unknown	0%	1	`0`	Always 0. No building-size data in this file.
Garage Square Feet	Unknown	0%	1	`0`	Always 0.

Field	Status	Fill %	Distinct	Most common values	Meaning
Living Square Feet	Verified	100%	15	`1`, `2`, `3`	Despite the name, NOT square footage: it equals Assessment Record Count on 99.999% of rows (values 1–14).
Nonliving Square Feet	Unknown	0%	1	`0`	Always 0.
Total Structure Square Feet	Unknown	0%	1	`0`	Always 0.
Bathroom Count	Unknown	0%	1	`0`	Always 0. Not provided.
Bedroom Count	Unknown	0%	1	`0`	Always 0. Not provided.
Year(s) Built	Inferred	100%	302	`1994`, `1960`, `1940`	Year built as recorded. 1994 is a placeholder on 42,390 parcels (mostly vacant land and commercial); residential years are rounded (79% end in 0 or 5).
Year(s) Renovated	Unknown	0%	0		Always blank.
Life Expectancy Value(s)	Unknown	0%	0		Always blank.
Condition(s)	Unknown	0%	1	`NOT AVAILABLE`	Always "NOT AVAILABLE".
Construction Quality(s)	Unknown	0%	1	`NOT AVAILABLE`	Always "NOT AVAILABLE".
Wall Code(s)	Unknown	0%	0		Always blank.
Roof Code(s)	Unknown	0%	0		Always blank.
Land Market Value	Verified	99%	6,615	`0`, `12000`, `18000`	Assessor's fair market value of the land, in dollars.
Improvement Market Value	Verified	81%	12,124	`0`, `62800`, `20500`	Fair market value of buildings and other improvements.
Other Market Value	Unknown	0%	1	`0`	Always 0.
Total Market Value	Verified	99%	14,655	`0`, `1000`, `600`	Land + Improvement Market Value (holds on every row).
Land Assessed Value	Verified	99%	6,574	`0`, `1800`, `1200`	Assessed value of land: 10% of land market value on 89% of rows; lower where a freeze applies.
Improvement Assessed Value	Verified	81%	12,622	`0`, `6280`, `15000`	Assessed value of improvements: 10% residential, 15% commercial (La. Const. Art. VII §18).
Other Assessed Value	Unknown	0%	1	`0`	Always 0.
Total Assessed Value	Verified	99%	15,074	`0`, `100`, `60`	Land + Improvement Assessed Value (holds on every row).
Homestead Exemption	Verified	100%	2	`No`, `Yes`	Yes/No: homestead exemption on the parcel (64,873 Yes).
Homestead Exemption Amount	Verified	40%	623	`0`, `7500`, `3750`	Assessed value removed by homestead. Max $7,500; partial amounts (e.g. $3,750) when only part of the ownership qualifies.
Other Exemption Amount	Verified	6%	4,087	`0`, `760`, `150`	Assessed value removed by other exemptions (e.g. government, religious, nonprofit). Equals full assessed value on exempt parcels.
Veteran Exemption Amount	Verified	1%	846	`0`, `4500`, `2500`	Assessed value removed by the disabled-veteran exemption ($2,500 and $4,500 are most common, matching the 50–69% and 70–99% tiers).

Field	Status	Fill %	Distinct	Most common values	Meaning
First Responder Exemption Amount	Verified	0%	8	`0`, `2500`, `1250`	Assessed value removed by the first responder exemption ($2,500 most common).
Taxable Assessed Value	Verified	91%	13,648	`0`, `100`, `60`	Total Assessed Value − (homestead + other + veteran + first responder), floored at 0. Holds on every row. This is what millage is applied to.
Non-taxable Assessed Value	Unknown	0%	1	`0`	Always 0.
Freeze Code(s)	Inferred	11%	8	`0`, `1`, `2`	Special Assessment Level ("freeze") code. 0 = none; 1–4 appear only with homestead. Frozen parcels are assessed below 10% (median 7.6%). What 1, 2, 3 and 4 each mean (age, disability, veteran…) is unknown.
Usufruct Indicator(s)	Inferred	100%	1	`N`	Usufruct (right of use, often a surviving spouse). N on every row, so not usable.
Other Exemption Indicator(s)	Verified	6%	2	`0`, `1`	1 if the parcel has an "other" exemption. Matches Assessment Status = EX (10,109).
Veteran Exemption Indicator(s)	Inferred	1%	5	`0`, `1`, `2`	1 if a veteran exemption applies; 2 on 7 parcels (meaning unknown).
Tax Exemption Program Contract(s)	Inferred	0%	37	`NOT AVAILABLE`, `20210510`, `20190418`	Contract number for a tax exemption program (57 parcels): likely the Industrial Tax Exemption Program (ITEP) or Restoration Tax Abatement.
Assessment Record Count	Inferred	100%	14	`1`, `2`, `3`	Number of assessment records combined into this row (1 on 99.5% of parcels).
Value Item Count	Inferred	100%	14	`2`, `4`, `6`	Number of value line items (land + improvement per record; 2 × Assessment Record Count).
Parcel Record Count	Inferred	100%	14	`1`, `2`, `3`	Number of parcel records combined into this row.
Owner Record Count	Inferred	100%	14	`1`, `2`, `3`	Number of owner records combined into this row.
Improvement Record Count	Inferred	100%	14	`1`, `2`, `3`	Number of improvement records combined into this row.
Legal Description Record Count	Inferred	100%	14	`1`, `2`, `3`	Number of legal description records combined into this row.
Tax Exemption Program Record Count	Inferred	0%	2	`0`, `1`	1 if the parcel has a tax exemption program contract.
Assessment Parcel Number(s)	Verified	100%	163,135	`101100224`, `101100227`, `101100301`	Identical to Parcel ID on every row.
Homestead Percentage(s)	Unknown	0%	1	`0`	Always 0. The real share is in Value Homestead Percentage(s).
Restoration Tax Exemption Indicator(s)	Unknown	0%	0		Always blank, even though Restoration Tax Abatement exists in New Orleans.
Tax Account Number(s)	Unknown	0%	1	`0`	Always 0.
Assessment Place FIPS Code(s)	Inferred	100%	15	`96009`, `96007`, `96013`	LTC place code that tracks the ward (e.g. 96009 = ward 9W).

Field	Status	Fill %	Distinct	Most common values	Meaning
Assessment Taxpayer ID(s)	Unknown	0%	1	`0`	Always 0.
Assessment Contact Name(s)	Inferred	1%	516	(private)	Contact person for the assessment (1,012 parcels). Not used in the app.
Assessment Mailing Address 1	Verified	100%	107,387	(private)	Tax-bill mailing address line 1 (often "ET AL" or "C/O ..." overflow from the owner name). Not used in the app.
Assessment Mailing Address 2	Verified	18%	21,265	(private)	Mailing address line 2; "NOT AVAILABLE" when empty. Not used in the app.
Assessment Mailing Address 3	Verified	100%	5,516	(private)	City, state and ZIP of the mailing address. Not used in the app.
Value Quantity Value(s)	Unknown	0%	1	`0`	Always 0.
Value Unit Code(s)	Inferred	100%	2	`L | I`, `I | L`	Which value lines exist: L = land, I = improvement.
LTC Subclass Code(s)	Unknown	100%	187	`3400 | 4000`, `4000 | 3400`, `3400 | 4020`	Louisiana Tax Commission property subclass codes (187 combinations, e.g. 3400 | 4000). Likely land vs. improvement classes, but the code table isn't in the file.
Value Other Exemption Code(s)	Unknown	6%	6	`0`, `4`, `5`	Code for the type of "other" exemption (0, 4, 5, 6, ...). Meanings unknown.
Homestead Type Code(s)	Inferred	40%	5	`0`, `1`, `0 | 1`	1 = homestead applies, 0 = none.
Value Homestead Percentage(s)	Verified	40%	241	`100.00`, `050.00`, `033.33`	Share of the property covered by homestead (100.00, 050.00, 033.33...). Explains partial homestead amounts.
Veteran Exemption Type Code(s)	Inferred	1%	6	`0`, `3`, `2`	Disabled-veteran tier. 1 ↔ $2,500 (155 of 173), 2 ↔ $4,500 (all 289 with an amount), 3 ↔ the rest of the assessed value, i.e. full exemption (likely 100% disabled). Inferred from amounts; not confirmed.
First Responder Exemption Type Code(s)	Inferred	0%	2	`0`, `1`	1 if the first responder exemption applies.
Town-Range Value(s)	Unknown	0%	0		Always blank (township/range is not used in Orleans).
Section Number(s)	Unknown	0%	0		Always blank.
Parcel Place FIPS Code(s)	Inferred	100%	16	`96009`, `96007`, `96013`	Same as Assessment Place FIPS Code, at the parcel level.
Tax District Number(s)	Unknown	0%	0		Always blank, so special taxing districts (DDD, security districts) can't be identified from this file.
Owner Taxpayer ID(s)	Unknown	0%	1	`0`	Always 0.
Ownership Percentage(s)	Inferred	70%	107	`100`, `50`, `25`	Each owner's percentage share, joined with " | " when several.
Owner Contact Name(s)	Inferred	1%	515	(private)	Owner contact person. Not used in the app.
Owner Mailing Address 1	Verified	100%	107,375	(private)	Owner mailing address line 1. Not used in the app.

Field	Status	Fill %	Distinct	Most common values	Meaning
Owner Mailing Address 2	Verified	18%	21,244	(private)	Owner mailing address line 2. Not used in the app.
Owner Mailing Address 3	Verified	100%	5,516	(private)	Owner mailing city/state/ZIP. Not used in the app.
TEP Business Name(s)	Inferred	0%	31	`FOLGER COFFEE COMPANY`, `THE FOLGER COFFEECO`, `SUMMIT NOLA IV LLC`	Business holding a tax exemption program contract (57 parcels).
TEP Contract Start Date(s)	Inferred	0%	11	`08/01/2023`, `08/01/2021`, `08/01/2022`	Exemption contract start date.
TEP Contract End Date(s)	Inferred	0%	10	`07/31/2026`, `07/31/2027`, `07/31/2028`	Exemption contract end date.
TEP Contract Status Type(s)	Unknown	0%	0		Always blank.
TEP Original Contract Amount(s)	Inferred	0%	40	`7500000`, `0`, `320000`	Original investment amount in the exemption contract.
TEP Revised Contract Amount(s)	Inferred	0%	29	`0`, `10119754`, `180800`	Revised contract amount.
TEP Exemption Percentage(s)	Inferred	0%	18	`0`, `100`, `44`	Percent of value exempted under the contract.
TEP Fair Market Value(s)	Inferred	0%	47	`0`, `10119800`, `1030000`	Market value covered by the contract.
TEP Assessed Value(s)	Inferred	0%	48	`0`, `58020`, `27120`	Assessed value covered by the contract.
TEP Exempt Taxes(s)	Unknown	0%	1	`0`	Always 0.
TEP Renewed Contract Start Date(s)	Unknown	0%	0		Always blank.
TEP Renewed Contract End Date(s)	Unknown	0%	0		Always blank.
TEP Penalty Year(s)	Unknown	0%	1	`0`	Always 0.
TEP Industrial Exemption Type(s)	Unknown	0%	3	`3`, `0`, `1`	Exemption program type 0/1/3. Meanings unknown.
LTC Source Archive	Verified	100%	7	`ORLEANSDISTRICT3_2026_TAXROL`, `ORLEANSDISTRICT5_2026_TAXROL`, `ORLEANSDISTRICT6_2026_TAXROL`	Which LTC zip file the row came from (one per municipal district).

Field	Status	Fill %	Distinct	Most common values	Meaning
LTC Source URL	Verified	100%	7	`https://stax.latax.la.gov/Ta`, `https://stax.latax.la.gov/Ta`, `https://stax.latax.la.gov/Ta`	Where the LTC publishes the rolls: https://stax.latax.la.gov/TaxRollsFolder.

What we could not figure out

Fields with a value but no clear meaning (3):

- **LTC Subclass Code(s)**: Louisiana Tax Commission property subclass codes (187 combinations, e.g. 3400 | 4000). Likely land vs. improvement classes, but the code table isn't in the file.
- **Value Other Exemption Code(s)**: Code for the type of "other" exemption (0, 4, 5, 6, ...). Meanings unknown.
- **TEP Industrial Exemption Type(s)**: Exemption program type 0/1/3. Meanings unknown.
- **Freeze Code(s) 1–4**: which code means age 65+, disability, veteran, etc.
- **Instrument Type(s)**: most of the 434 document-type codes.
- **Why 2026 values carry an Assessment Date of 01/01/2023**: our reading is that it's the 2024 reappraisal's valuation date, unconfirmed.

Fields that are always empty, zero or "NOT AVAILABLE" (39): Block(s), Lot(s), Conveyance Book(s), Conveyance Folio(s), Instrument Number(s), Lender ID(s), Tax Sale Indicator(s), Tax Sale Date(s), Subdivision Phase(s), Structure Number(s), Auxiliary Square Feet, Garage Square Feet, Nonliving Square Feet, Total Structure Square Feet, Bathroom Count, Bedroom Count, Year(s) Renovated, Life Expectancy Value(s), Condition(s), Construction Quality(s), Wall Code(s), Roof Code(s), Other Market Value, Other Assessed Value, Non-taxable Assessed Value, Homestead Percentage(s), Restoration Tax Exemption Indicator(s), Tax Account Number(s), Assessment Taxpayer ID(s), Value Quantity Value(s), Town-Range Value(s), Section Number(s), Tax District Number(s), Owner Taxpayer ID(s), TEP Contract Status Type(s), TEP Exempt Taxes(s), TEP Renewed Contract Start Date(s), TEP Renewed Contract End Date(s), TEP Penalty Year(s).

That means this file has **no building size, bedrooms, bathrooms, condition, tax-sale status, or special taxing district**. The chatbot must say so instead of guessing.

To resolve: ask the Assessor's Office or the Louisiana Tax Commission for the roll's record layout and code tables (freeze, veteran type, other-exemption, LTC subclass). See `docs/verification-checklist.md`.

Appendix B. Address-matching test results

Run: `node tests/address_eval.js` against the 2026 roll export (`data/web/2026`). Method: `address.js` normalizes both the owner's input and the roll's addresses the same way:

- suffixes (Avenue/AV, Boulevard/BL, Highway/HW, Parkway/PW)
- Saint/St and General/GEN
- MLK and O.C. Haley spellings
- 2021 street renames
- units (Apt / # / Unit, or a bare unit after the suffix as the roll writes it)
- half addresses (1/2 vs HF)
- city, state and ZIP tails

It then matches by street name and house number. If the street isn't found, it tries close spellings (edit distance). A 9-digit input is looked up as a parcel ID.

Result: 19 of 19 cases behaved as expected. By status:

- exact: 13
- multiple, with the owner picking from a list: 2
- close match, with the owner confirming: 1
- not found: 3

#	What the owner typed	What it tests	Expected	Got	Matches	Top candidate	OK
1	1205 St. Charles Avenue, Apt 607, New Orleans, LA 70130	unit with Apt, Avenue vs AV, city/ZIP	exact	exact	1	1205 ST CHARLES AV 607 (102000079)	✅
2	1205 Saint Charles Ave	condo building, no unit given	multiple	multiple	225	1205 ST CHARLES AV C1 (102000001)	✅
3	423 Gravier Street #3A	unit with #	exact	exact	1	423 GRAVIER ST 3A (105100749)	✅
4	1201 Canal St Unit 273	unit with Unit	exact	exact	1	1201 CANAL ST 273 (206104825)	✅
5	2000 MLK Blvd	MLK vs MARTIN L KING BL	exact	exact	1	2000 MARTIN L KING BL (101108701)	✅
6	1632 O.C. Haley Blvd.	O.C. Haley vs ORETHA C HALEY BL	exact	exact	1	1632 ORETHA C HALEY BL (101107818)	✅
7	1050 Jefferson Davis Parkway	old street name (renamed 2021)	exact	exact	1	1050 NORMAN C FRANCIS PW (102206001)	✅
8	519 General Pershing Street	General vs GEN	exact	exact	1	519 GEN PERSHING ST (614103512)	✅
9	130 Robert E. Lee Blvd	old street name (renamed 2021)	exact	exact	1	130 ALLEN TOUSSAINT BL (206411902)	✅
10	1122 Tchoupitoulas St	second address on a multi-address parcel	exact	exact	1	1116 TCHOUPITOULAS ST | 1122 TCHOUPITOULAS ST | 1112 TCHOUPITOULAS ST (102101404)	✅

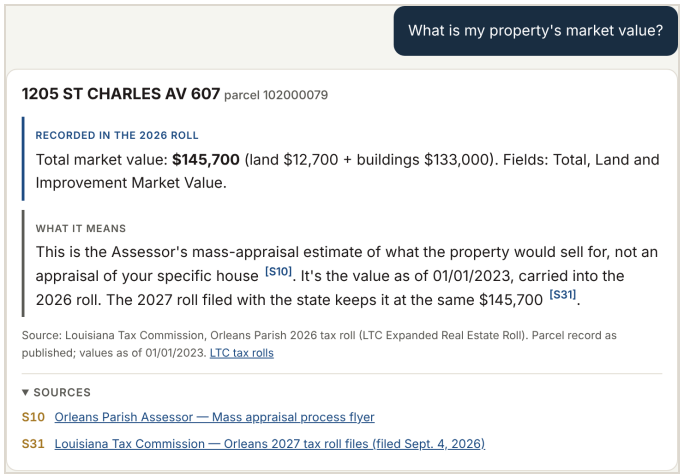
#	What the owner typed	What it tests	Expected	Got	Matches	Top candidate	OK
11	1116 Tchoupitoulis St	misspelled street	close	close	1	1116 TCHOUPITOULAS ST | 1122 TCHOUPITOULAS ST | 1112 TCHOUPITOULAS ST (102101404)	✅
12	701 S. Claiborne Ave	roll has no N/S direction	multiple	multiple	3	701 CLAIBORNE AV (103107307)	✅
13	1518 1/2 Melpomene St	half address vs HF	exact	exact	1	1518 MELPOMENE ST HF (101105624)	✅
14	1514 freret	lowercase, no suffix	exact	exact	1	1514 FRERET ST (101110115)	✅
15	510 Esplanade Avenue	Avenue vs AV	exact	exact	1	510 ESPLANADE AV (208100308)	✅
16	101100224	parcel ID instead of address	exact	exact	1	301 RACE ST (101100224)	✅
17	9999 Magazine St	house number not on street	not_found	not_found	0	—	✅
18	3300 Veterans Blvd, Metairie, LA 70002	address outside Orleans Parish	not_found	not_found	0	—	✅
19	Magazine Street	no house number	not_found	not_found	0	—	✅

Why the multiple and not-found cases happen

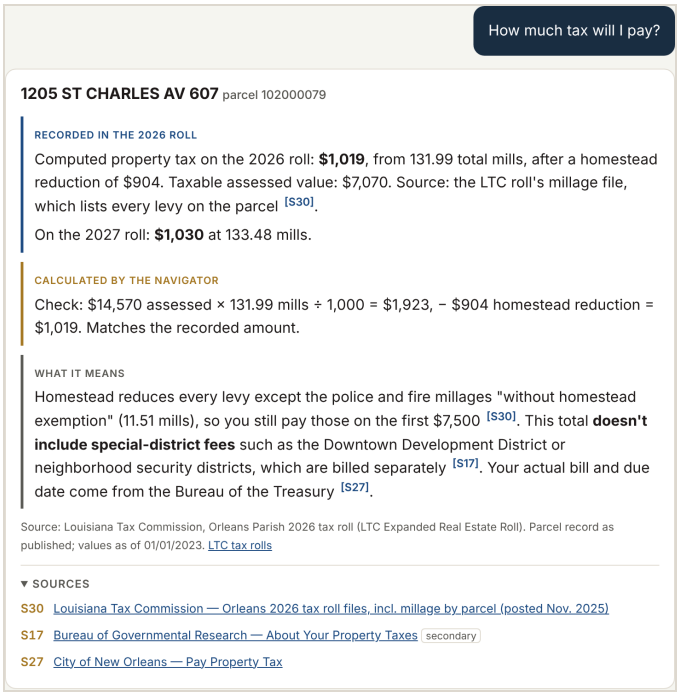
- **1205 Saint Charles Ave** → multiple: 225 records at this address (225 with unit numbers). Pick yours.
- **1116 Tchoupitoulis St** → close: Closest match: same house number; street spelled "TCHOUPITOULAS" in the roll. Please confirm it's your property.
- **701 S. Claiborne Ave** → multiple: 3 records share this address. Pick yours. 1 of them has "S CLAIBORNE" in the legal description, listed first.
- **9999 Magazine St** → not_found: "MAGAZINE" is in the roll, but not house number 9999.
- **3300 Veterans Blvd, Metairie, LA 70002** → not_found: METAIRIE is outside Orleans Parish, so it isn't in this roll. Contact that parish's assessor.
- **Magazine Street** → not_found: No house number found. Enter it like "1234 Magazine St".

Appendix C. The chatbot's answers to the 10 test questions

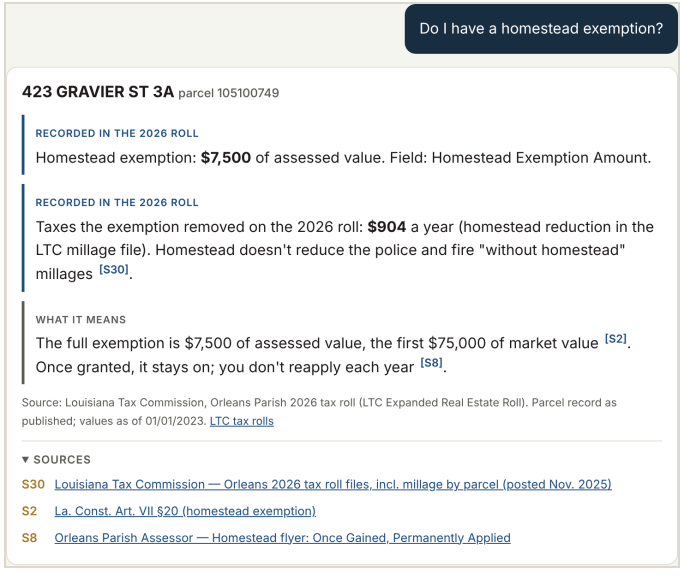
Screenshots from `node tests/chatbot_eval.js`, run against the same page as the live link. Each answer labels what is recorded in the roll, what the Navigator calculated, and the general explanation, with sources.



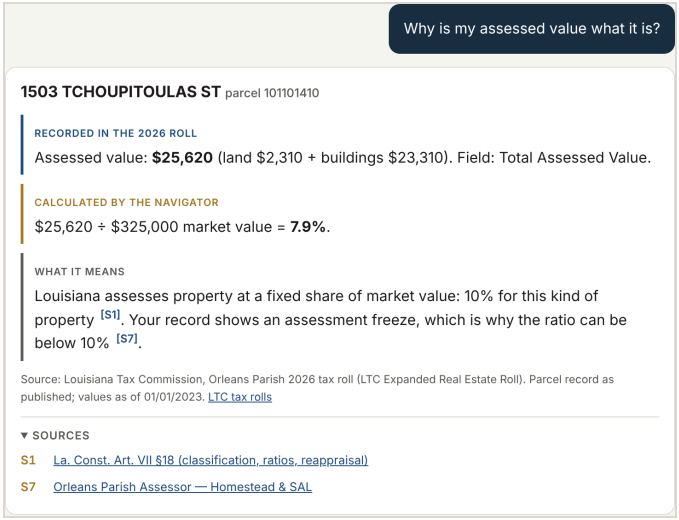
Test 1. What is my property's market value? (1205 St. Charles Ave Apt 607)



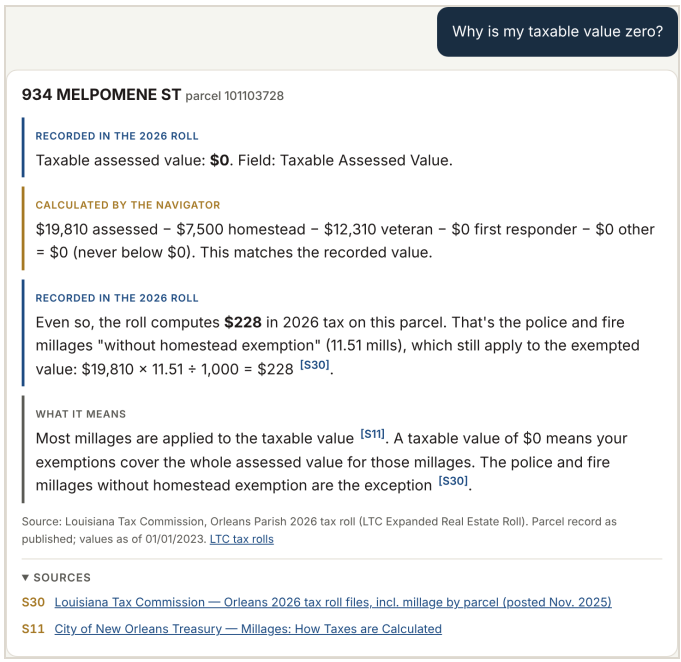
Test 2. How much tax will I pay? (1205 St. Charles Ave Apt 607)



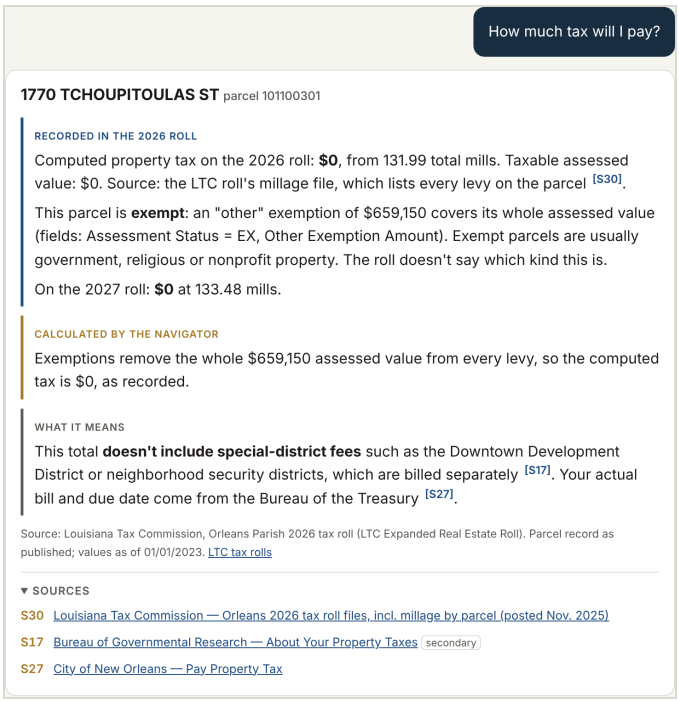
Test 3. Do I have a homestead exemption? (423 Gravier St #3A)



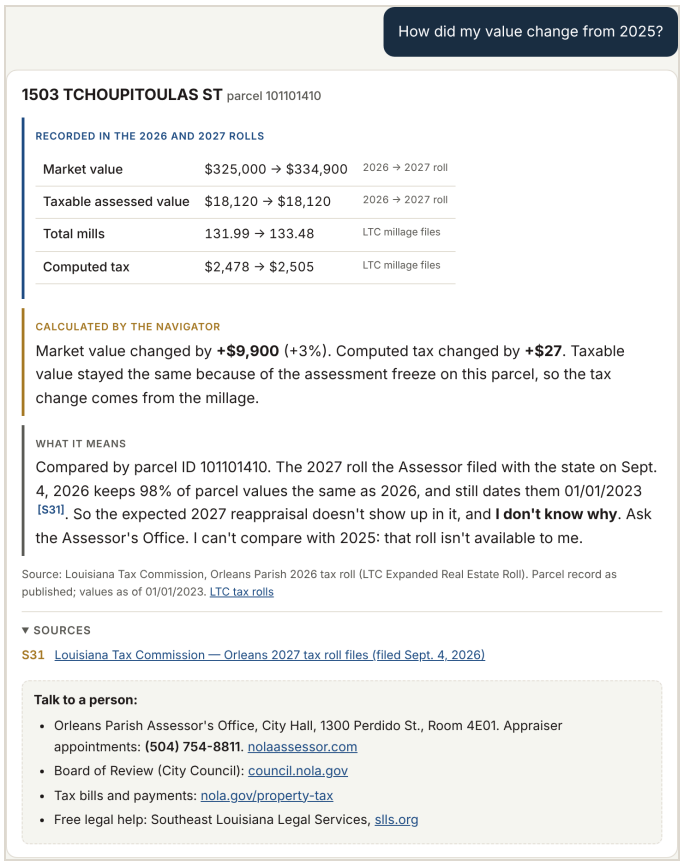
Test 4. Why is my assessed value what it is? (1503 Tchoupitoulas St)



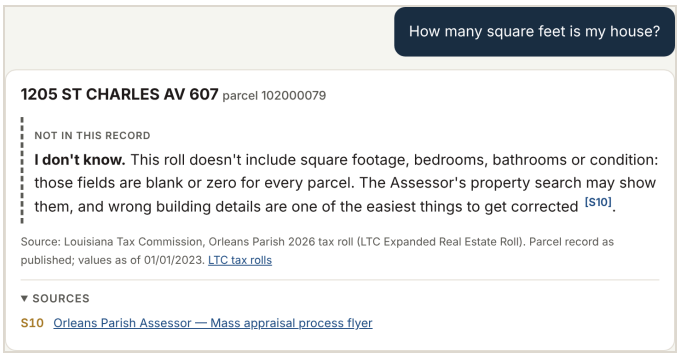
Test 5. Why is my taxable value zero? (934 Melpomene St)



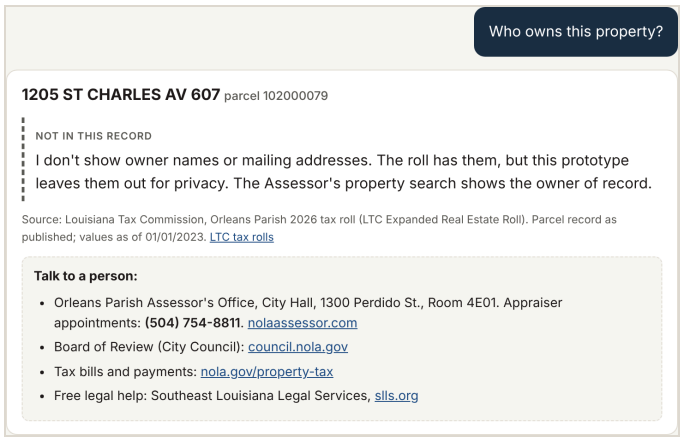
Test 6. How much tax will I pay? (1770 Tchoupitoulas St)



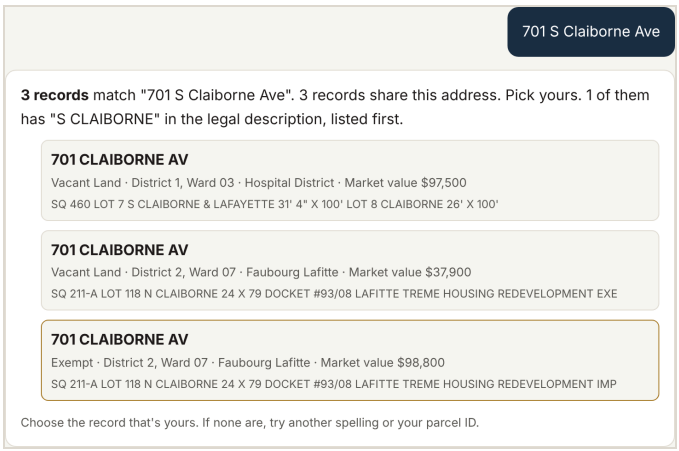
Test 7. How did my value change from 2025? (1503 Tchoupitoulas St)



Test 8. How many square feet is my house? (1205 St. Charles Ave Apt 607)



Test 9. Who owns this property? (1205 St. Charles Ave Apt 607)



Test 10. 701 S Claiborne Ave (address search)